

Part I – Short Answer (56 pts)

(-1) A (-2)
B (-3)
C (-5)
D (-7)

1. Data Structures in the Wild (16 pts)

You are maintaining data from a dachshund competition. Each row contains:

- name
- score
- coat

Before traversing the dataset row-by-row, you will choose a data structure that **BEST** supports each task. Then assume you process a dataset with n rows, one row at a time. For each scenario below:

- choose the best data structure
- give the overall Big-O runtime for processing all n rows
- briefly justify your answer

(a) Track how many dogs have the same coat.

Data Structure: Dictionary

Big-O: $O(n)$

Justification:

n elements & lookups with key is $O(1)$

Big O - of (a) the operation: $O(1)$

(b) Check whether a dog name has already appeared.

Data Structure: Set

Big-O: $O(n)$

Justification:

n rows and containment is on average $O(1)$

Big-O of (b) the operation: $O(1)$

(c) Store scores in order, and support efficient updates as new competition data is added weekly.

Data Structure: BST

Big-O: $O(n \log n)$

Justification:

or sorted list (linked list best bc of skimming)
sort BEST method: $O(n \log n)$

insert n elements into BST

Big-O of (c) the operation: $O(n)$ on sorted list

(d) Repeatedly retrieve the highest score efficiently. $O(\log n)$ on BST

Data Structure: Heap

Big-O: $n \log n$

Justification:

n touch all the rows then $\log n$ for insert

Big O - of (d) the operation (1)

2. Pandas (6 pts)

```
df = pd.DataFrame({
    "city": ["Memphis", "Memphis", "Nashville", "Oxford", "Memphis"],
    "score": [91, 85, 95, 88, 92],
    "year": [2024, 2024, 2025, 2025, 2025]})
```